

Dionex_COMPASS_November2022

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Daily Set up

Read in and Format the raw data - change wd & file names (Make sure files are UTF-8 encoded)

```
# SULFATE DATA:

## Read in raw data file from Dionex - copied and saved as a txt
Sdat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202211_S04.txt", sep='\t' , header=T, skip=3)
head(Sdat)

##      X      X.1      X.2 IC.S04 IC.S04.1 IC.S04.2 IC.S04.3 IC.S04.4
## 1 1 Lab Blank      Unknown  n.a.      n.a.      n.a.      n.a.      n.a.
## 2 2 Lab Blank      Unknown  n.a.      n.a.      n.a.      n.a.      n.a.
## 3 3 Standard 1      Unknown  4.397    0.5017    5.85    0.4070    2.53
## 4 4 Blank          Unknown  n.a.      n.a.      n.a.      n.a.      n.a.
## 5 5 Standard 1 Calibration Standard 4.393    0.5034    5.87    0.4084    2.54
## 6 6 Standard 2 Calibration Standard 4.400    1.0993    6.12    0.8917    5.39

##Fix Sample IDs that were entered incorrectly
Sdat <- Sdat %>%
  mutate(X.1 = ifelse(X == "192", paste0(X.1, "_dup"), X.1)) %>%
  mutate(X.1 = ifelse(X == "163", paste0(X.1, "_spk"), X.1))

## Only keep the columns that we need
Sdat <- Sdat[ ,c(2,5)] # dont need this here
head(Sdat)

##      X.1 IC.S04.1
## 1 Lab Blank      n.a.
## 2 Lab Blank      n.a.
## 3 Standard 1 0.5017
## 4 Blank      n.a.
## 5 Standard 1 0.5034
## 6 Standard 2 1.0993

## Name the columns correctly
colnames(Sdat) <- c("Sample_ID", "S04_ppm")
Sdat$Sample_ID <- as.factor(Sdat$Sample_ID)
Sdat$S04_ppm <- as.numeric(Sdat$S04_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Sdat <- as.data.frame(Sdat)
head(Sdat)
```

```
##      Sample_ID S04_ppm
## 1 Lab Blank      NA
## 2 Lab Blank      NA
## 3 Standard 1  0.5017
## 4      Blank      NA
## 5 Standard 1  0.5034
## 6 Standard 2  1.0993
```

```
#Chloride data
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Cldat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202211_Cl.txt", sep='\t' , header=T, skip=3)
head(Cldat)
```

```
##      X      X.1      X.2 IC.Cl IC.Cl.1 IC.Cl.2 IC.Cl.3 IC.Cl.4
## 1 1 Lab Blank      Unknown 3.270  0.0241 100.00  0.0271  0.19
## 2 2 Lab Blank      Unknown 3.270  0.0251 100.00  0.0282  0.20
## 3 3 Standard 1      Unknown 3.273  5.7878  93.58  6.5054  43.94
## 4 4      Blank      Unknown 3.270  0.0260 100.00  0.0292  0.21
## 5 5 Standard 1 Calibration Standard 3.273  5.7959  93.56  6.5145  44.12
## 6 6 Standard 2 Calibration Standard 3.273 12.0880  93.22 13.5866  90.14
```

```
##Fix Sample IDs that were entered incorrectly
```

```
Cldat <- Cldat %>%
  mutate(X.1 = ifelse(X == "192", paste0(X.1, "_dup"), X.1)) %>%
  mutate(X.1 = ifelse(X == "163", paste0(X.1, "_spk"), X.1))
```

```
## Only keep the columns that we need
```

```
Cldat <- Cldat[,c(2,5)]
head(Cldat)
```

```
##      X.1 IC.Cl.1
## 1 Lab Blank 0.0241
## 2 Lab Blank 0.0251
## 3 Standard 1 5.7878
## 4      Blank 0.0260
## 5 Standard 1 5.7959
## 6 Standard 2 12.0880
```

```
## Name the columns correctly
```

```
colnames(Cldat) <- c("Sample_ID", "Cl_ppm")
Cldat$Sample_ID <- as.factor(Cldat$Sample_ID)
Cldat$Cl_ppm <- as.numeric(Cldat$Cl_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Cldat <- as.data.frame(Cldat)
head(Cldat)
```

```
##      Sample_ID  Cl_ppm
## 1  Lab Blank  0.0241
## 2  Lab Blank  0.0251
## 3 Standard 1  5.7878
## 4      Blank  0.0260
## 5 Standard 1  5.7959
## 6 Standard 2 12.0880
```

```
## Bring the data back together:
all_dat <- merge(Sdat, Cldat, by="Sample_ID")
head(all_dat)
```

```
##      Sample_ID  S04_ppm      Cl_ppm
## 1           2544.4789 19269.7855
## 2           2544.4789  5007.1144
## 3           2544.4789   0.0179
## 4           2544.4789  5111.7920
## 5           2544.4789        NA
## 6           784.5324 19269.7855
```

```
## Remove empty lines
all_dat <- all_dat[!(is.na(all_dat$Sample_ID) | all_dat$Sample_ID==""), ]
head(all_dat)
```

```
##              Sample_ID  S04_ppm  Cl_ppm
## 26      1_202211_MSM_UP_LysA_10cm  264.6481 3270.8644
## 27     10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636
## 28    10_202211_MSM_TR_LysA_20cm_spk 1248.1115 6395.9330
## 29    100_202211_GCrew_TR_LysC_45cm   19.3955  133.2425
## 30 100_202211_GCrew_TR_LysC_45cm_spk  158.7476  134.0034
## 31    101_202211_GCrew_WC_SipA_10cm  705.7256 6762.9137
```

```
all_dat <- droplevels(all_dat[!all_dat$Sample_ID == 'Lab Blank',])
head(all_dat)
```

```
##              Sample_ID  S04_ppm  Cl_ppm
## 26      1_202211_MSM_UP_LysA_10cm  264.6481 3270.8644
## 27     10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636
## 28    10_202211_MSM_TR_LysA_20cm_spk 1248.1115 6395.9330
## 29    100_202211_GCrew_TR_LysC_45cm   19.3955  133.2425
## 30 100_202211_GCrew_TR_LysC_45cm_spk  158.7476  134.0034
## 31    101_202211_GCrew_WC_SipA_10cm  705.7256 6762.9137
```

Fix Sample IDs

```
##Remove samples that were ran several times and had different values
all_dat <- all_dat %>%
  filter(!str_detect(Sample_ID, "202211_MSM_TR_LysB"))
```

Pull out standards - could do some checks here if we want

```
stds <- all_dat[grepl("Standard", all_dat$Sample_ID),]
head(stds)
```

```
##      Sample_ID S04_ppm Cl_ppm
## 174 Standard 1  0.5034 5.7959
## 175 Standard 1  0.5034 5.8500
## 176 Standard 1  0.5034 5.8551
## 177 Standard 1  0.5034 5.9283
## 178 Standard 1  0.5034 5.9000
## 179 Standard 1  0.5034 5.7635
```

```
stds_chk_S <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(S04_ppm), sd = sd(S04_ppm))
stds_chk_S$cv <- (stds_chk_S$sd/stds_chk_S$mean)*100
stds_chk_S$flag <- ifelse(stds_chk_S$cv < 2, 'YES', 'NO, rerun')
head(stds_chk_S)
```

```
## # A tibble: 5 x 5
##   Sample_ID mean    sd    cv flag
##   <fct>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  0.506 0.0109 2.16 NO, rerun
## 2 Standard 2  1.11  0.0162 1.46 YES
## 3 Standard 3  2.22  0.0416 1.88 YES
## 4 Standard 4  8.80  0.100  1.14 YES
## 5 Standard 5 20.4  0.190  0.935 YES
```

```
stds_chk_Cl <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(Cl_ppm), sd = sd(Cl_ppm))
stds_chk_Cl$cv <- (stds_chk_Cl$sd/stds_chk_Cl$mean)*100
stds_chk_Cl$flag <- ifelse(stds_chk_Cl$cv < 2, 'YES', 'NO, rerun')
head(stds_chk_Cl)
```

```
## # A tibble: 5 x 5
##   Sample_ID mean    sd    cv flag
##   <fct>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  5.84  0.0570 0.975 YES
## 2 Standard 2 12.2  0.0665 0.546 YES
## 3 Standard 3 23.4  0.186  0.796 YES
## 4 Standard 4 89.3  0.628  0.704 YES
## 5 Standard 5 203.  1.53   0.753 YES
```

Calculate mmol/L concentrations

```
#remove standards from sample dataframe
sampledat <- all_dat[!grepl("Standard", all_dat$Sample_ID),]
head(sampledat)
```

```
##               Sample_ID  S04_ppm  Cl_ppm
## 1      1_202211_MSM_UP_LysA_10cm 264.6481 3270.8644
## 2      10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636
## 3      10_202211_MSM_TR_LysA_20cm_spk 1248.1115 6395.9330
## 4      100_202211_GCrew_TR_LysC_45cm 19.3955 133.2425
## 5 100_202211_GCrew_TR_LysC_45cm_spk 158.7476 134.0034
## 6      101_202211_GCrew_WC_SipA_10cm 705.7256 6762.9137
```

```
# Constants needed for calculations:
clmw <- 35.45      #molecular weight of Chloride: 35.45
smw <- 32.06      #molecular weight of sulfur: 32.06

# Convert ppm to mmol/L
sampledat$S04_mM <- (sampledat$S04_ppm / smw)
sampledat$Cl_mM <- (sampledat$Cl_ppm / clmw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
sampledat$salinity <- ((1.8070 * sampledat$Cl_ppm) + 0.026) / 1000

head(sampledat)
```

```
##               Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1      1_202211_MSM_UP_LysA_10cm 264.6481 3270.8644  8.254775 92.266979
## 2      10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636 41.341089 193.951018
## 3      10_202211_MSM_TR_LysA_20cm_spk 1248.1115 6395.9330 38.930490 180.421241
## 4      100_202211_GCrew_TR_LysC_45cm 19.3955 133.2425  0.604975  3.758604
## 5 100_202211_GCrew_TR_LysC_45cm_spk 158.7476 134.0034  4.951578  3.780068
## 6      101_202211_GCrew_WC_SipA_10cm 705.7256 6762.9137 22.012651 190.773306
##      salinity
## 1  5.9104780
## 2 12.4241694
## 3 11.5574769
## 4  0.2407952
## 5  0.2421701
## 6 12.2206111
```

Pull out dups and check with percent difference

```
#Show me the data that we have from the calculations  
head(sampledat)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM  
## 1      1_202211_MSM_UP_LysA_10cm 264.6481 3270.8644 8.254775 92.266979  
## 2      10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636 41.341089 193.951018  
## 3      10_202211_MSM_TR_LysA_20cm_spk 1248.1115 6395.9330 38.930490 180.421241  
## 4      100_202211_GCrew_TR_LysC_45cm 19.3955 133.2425 0.604975 3.758604  
## 5      100_202211_GCrew_TR_LysC_45cm_spk 158.7476 134.0034 4.951578 3.780068  
## 6      101_202211_GCrew_WC_SipA_10cm 705.7256 6762.9137 22.012651 190.773306  
##      salinity  
## 1      5.9104780  
## 2     12.4241694  
## 3     11.5574769  
## 4      0.2407952  
## 5      0.2421701  
## 6     12.2206111
```

```
#pull out any rows that have "dup" in the SampleID column  
dups <- sampledat %>%  
  filter(str_detect(Sample_ID, "dup")) #have to change this to match data  
head(dups)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM  
## 1 109_202211_GCrew_WC_SipC_45cm_dup 490.3857 5847.447 15.29587 164.9491  
## 2 119_202211_GWI_UP_LysC_20cm_dup 1062.6500 8633.381 33.14566 243.5368  
## 3 129_202211_GWI_WC_SIPa_10cm_dup 2192.9807 15519.800 68.40239 437.7941  
## 4      139_202211_GWI_SW_C_dup 1972.7418 14364.361 61.53281 405.2006  
## 5      19_202211_MSM_WC_SipA_20cm_dup 2527.3082 14813.024 78.83057 417.8568  
## 6      29_202211_MSM_SW_C_dup 2245.2770 11988.478 70.03359 338.1799  
##      salinity  
## 1     10.56636  
## 2     15.60055  
## 3     28.04430  
## 4     25.95643  
## 5     26.76716  
## 6     21.66321
```

```
#remove these from sample dataframe in a new dataframe  
sampledat2 <- sampledat %>%  
  filter(!str_detect(Sample_ID, "dup")) %>%  
  filter(!str_detect(Sample_ID, "spk"))  
head(sampledat2)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM  
## 1      1_202211_MSM_UP_LysA_10cm 264.6481 3270.8644 8.254775 92.266979  
## 2      10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636 41.341089 193.951018  
## 3      100_202211_GCrew_TR_LysC_45cm 19.3955 133.2425 0.604975 3.758604  
## 4      101_202211_GCrew_WC_SipA_10cm 705.7256 6762.9137 22.012651 190.773306
```

```
## 5 102_202211_GCrew_WC_SipA_20cm 241.4455 6041.3935 7.531051 170.420127
## 6 103_202211_GCrew_WC_SipA_45cm 630.0428 6834.2797 19.651990 192.786451
##      salinity
## 1 5.9104780
## 2 12.4241694
## 3 0.2407952
## 4 12.2206111
## 5 10.9168241
## 6 12.3495694
```

```
#remove the dup from these IDs so we will have duplicate sample names
dups$Sample_ID<-gsub("_dup","",as.character(dups$Sample_ID))
dups <- dups[ , -c(2,3)]
colnames(dups) <- c('Sample_ID', 'SO4_mM_dup', 'Cl_mM_dup', 'salinity_dup')
head(dups)
```

```
##              Sample_ID SO4_mM_dup Cl_mM_dup salinity_dup
## 1 109_202211_GCrew_WC_SipC_45cm 15.29587 164.9491 10.56636
## 2 119_202211_GWI_UP_LysC_20cm 33.14566 243.5368 15.60055
## 3 129_202211_GWI_WC_SipA_10cm 68.40239 437.7941 28.04430
## 4 139_202211_GWI_SW_C 61.53281 405.2006 25.95643
## 5 19_202211_MSM_WC_SipA_20cm 78.83057 417.8568 26.76716
## 6 29_202211_MSM_SW_C 70.03359 338.1799 21.66321
```

```
#put it back together with the old data set and look for duplicates
QAdups <- merge(sampldat2, dups)
head(QAdups)
```

```
##              Sample_ID SO4_ppm Cl_ppm SO4_mM Cl_mM salinity
## 1 109_202211_GCrew_WC_SipC_45cm 467.7261 5581.663 14.58909 157.4517 10.086092
## 2 119_202211_GWI_UP_LysC_20cm 1144.9459 9289.281 35.71260 262.0390 16.785757
## 3 139_202211_GWI_SW_C 2131.0356 15474.590 66.47023 436.5188 27.962610
## 4 19_202211_MSM_WC_SipA_20cm 2365.1949 14059.697 73.77401 396.6064 25.405898
## 5 29_202211_MSM_SW_C 2098.9347 12472.348 65.46896 351.8293 22.537559
## 6 39_202211_MSM_RHZ_TR_SF_2 713.7607 5163.679 22.26328 145.6609 9.330794
## SO4_mM_dup Cl_mM_dup salinity_dup
## 1 15.29587 164.9491 10.56636
## 2 33.14566 243.5368 15.60055
## 3 61.53281 405.2006 25.95643
## 4 78.83057 417.8568 26.76716
## 5 70.03359 338.1799 21.66321
## 6 20.53351 136.7923 8.76269
```

```
QAdups$SO4_dups_chk <- ((abs(QAdups$SO4_mM-QAdups$SO4_mM_dup))/((QAdups$SO4_mM+QAdups$SO4_mM_dup)/2))*100
QAdups$SO4_dups_flag <- ifelse(QAdups$SO4_dups_chk <10, 'YES', 'NO, rerun')

QAdups$Cl_dups_chk <- ((abs(QAdups$Cl_mM-QAdups$Cl_mM_dup))/((QAdups$Cl_mM+QAdups$Cl_mM_dup)/2))*100
QAdups$Cl_dups_flag <- ifelse(QAdups$Cl_dups_chk <10, 'YES', 'NO, rerun')

head(QAdups)
```

```
##              Sample_ID SO4_ppm Cl_ppm SO4_mM Cl_mM salinity
```

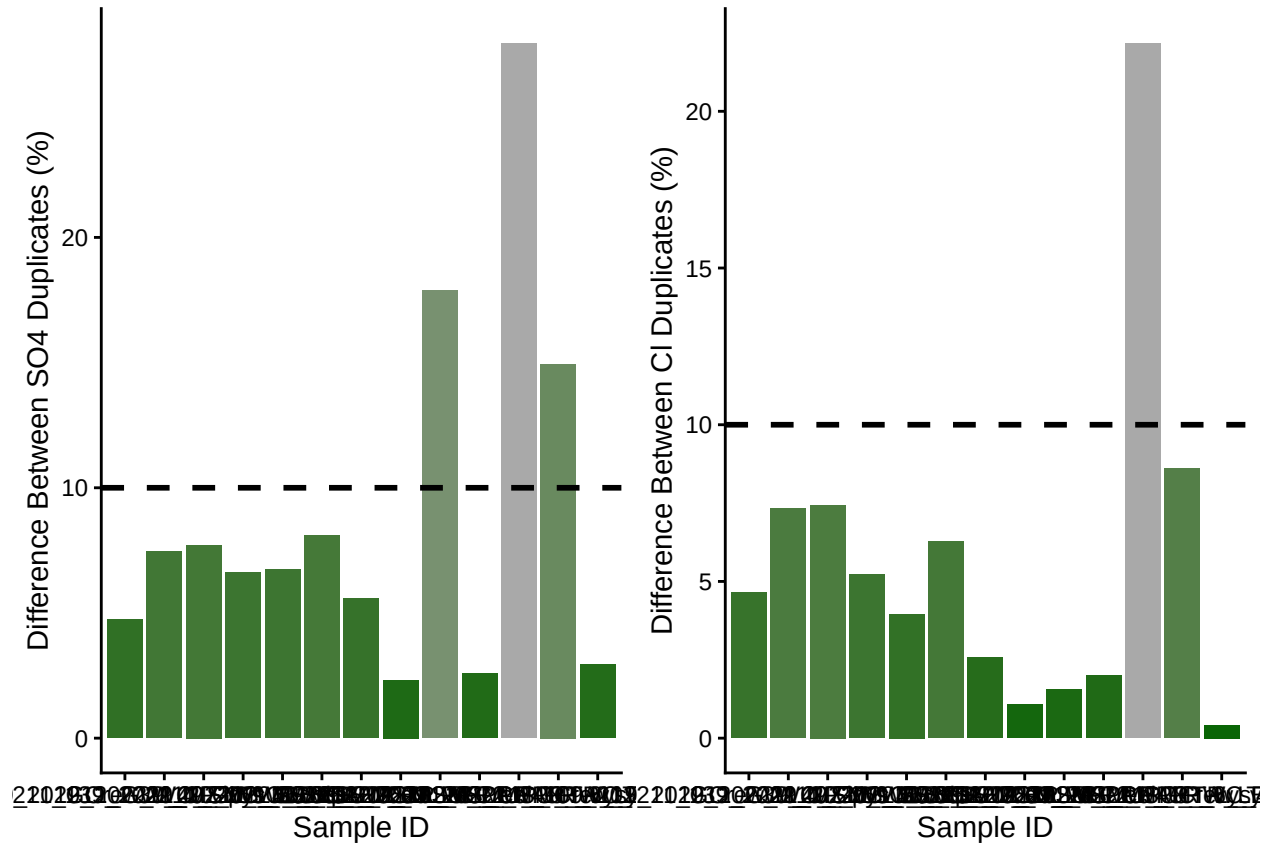
```
## 1 109_202211_GCrew_WC_SipC_45cm 467.7261 5581.663 14.58909 157.4517 10.086092
## 2 119_202211_GWI_UP_LysC_20cm 1144.9459 9289.281 35.71260 262.0390 16.785757
## 3 139_202211_GWI_SW_C 2131.0356 15474.590 66.47023 436.5188 27.962610
## 4 19_202211_MSM_WC_SipA_20cm 2365.1949 14059.697 73.77401 396.6064 25.405898
## 5 29_202211_MSM_SW_C 2098.9347 12472.348 65.46896 351.8293 22.537559
## 6 39_202211_MSM_RHZ_TR_SF_2 713.7607 5163.679 22.26328 145.6609 9.330794
## S04_mM_dup Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag Cl_dups_chk
## 1 15.29587 164.9491 10.56636 4.730053 YES 4.650997
## 2 33.14566 243.5368 15.60055 7.455703 YES 7.319232
## 3 61.53281 405.2006 25.95643 7.714541 YES 7.441473
## 4 78.83057 417.8568 26.76716 6.627009 YES 5.218266
## 5 70.03359 338.1799 21.66321 6.737347 YES 3.956287
## 6 20.53351 136.7923 8.76269 8.083640 YES 6.279669
## Cl_dups_flag
## 1 YES
## 2 YES
## 3 YES
## 4 YES
## 5 YES
## 6 YES
```

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_chk, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar, ncol=2, nrow=1)
```

```
#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(SO4_dups_flag) %>%
  summarise(S_no_rows = length(SO4_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(CI_dups_flag) %>%
  summarise(CI_no_rows = length(CI_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "CI_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$SO4_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$CI_Percent <- (Perc_dups2$CI_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##      Flag S_no_rows      Flag CI_no_rows Total S_Percent CI_Percent
## 1 NO, rerun      3 NO, rerun      1    13  23.07692   7.692308
## 2      YES     10      YES      12    13  76.92308  92.307692
```

Pull out dups and check with cv

```

#the cv
# calculate the sd bewteen the two duplicates then divide by the mean and multiply by 100

df2 <- as.data.frame(QAdups$S04_mM)
df2$dups <- QAdups$S04_mM_dup

df2$sds <- apply(df2,1,sd)

QAdups$S04_dups_cv <- (df2$sds)/((QAdups$S04_mM+QAdups$S04_mM_dup)/2) * 100
QAdups$S04_dups_cv_flag <- ifelse(QAdups$S04_dups_cv <11, 'YES', 'NO, rerun')

head(QAdups)

```

```

##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM   salinity
## 1 109_202211_GCrew_WC_SipC_45cm 467.7261 5581.663 14.58909 157.4517 10.086092
## 2  119_202211_GWI_UP_LysC_20cm 1144.9459 9289.281 35.71260 262.0390 16.785757
## 3           139_202211_GWI_SW_C 2131.0356 15474.590 66.47023 436.5188 27.962610
## 4   19_202211_MSM_WC_SipA_20cm 2365.1949 14059.697 73.77401 396.6064 25.405898
## 5           29_202211_MSM_SW_C 2098.9347 12472.348 65.46896 351.8293 22.537559
## 6   39_202211_MSM_RHZ_TR_SF_2  713.7607  5163.679 22.26328 145.6609  9.330794
##   S04_mM_dup Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag Cl_dups_chk
## 1  15.29587 164.9491   10.56636    4.730053         YES    4.650997
## 2  33.14566 243.5368   15.60055    7.455703         YES    7.319232
## 3  61.53281 405.2006   25.95643    7.714541         YES    7.441473
## 4  78.83057 417.8568   26.76716    6.627009         YES    5.218266
## 5  70.03359 338.1799   21.66321    6.737347         YES    3.956287
## 6  20.53351 136.7923    8.76269    8.083640         YES    6.279669
##   Cl_dups_flag S04_dups_cv S04_dups_cv_flag
## 1           YES    3.344653             YES
## 2           YES    5.271978             YES
## 3           YES    5.455004             YES
## 4           YES    4.686003             YES
## 5           YES    4.764023             YES
## 6           YES    5.715997             YES

```

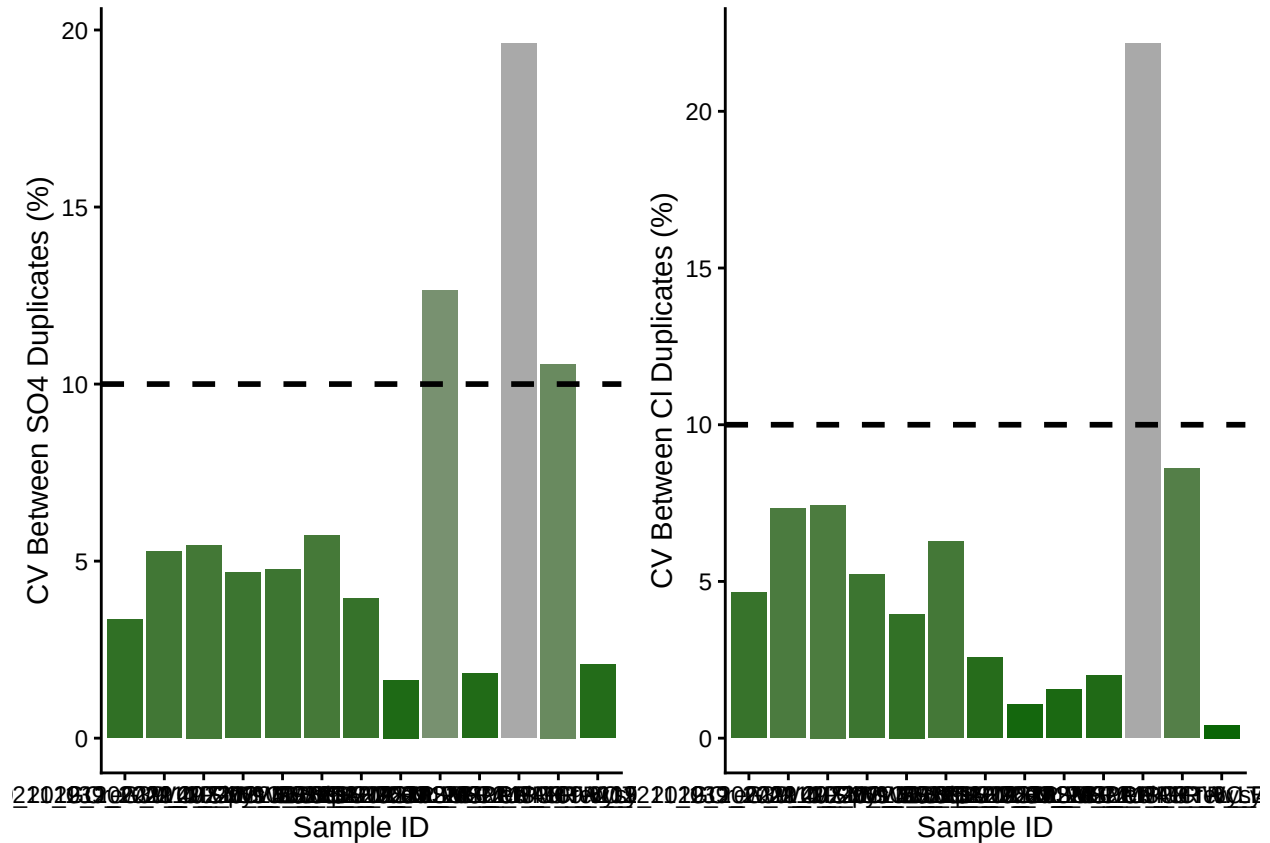
```

#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_cv, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar,ncol=2, nrow=1)

```



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(S04_dups_flag) %>%
  summarise(S_no_rows = length(S04_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(C1_dups_flag) %>%
  summarise(C1_no_rows = length(C1_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "C1_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$S04_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$C1_Percent <- (Perc_dups2$C1_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##      Flag S_no_rows      Flag C1_no_rows Total S_Percent C1_Percent
## 1 NO, rerun      3 NO, rerun      1     13  23.07692   7.692308
## 2      YES      10      YES      12     13  76.92308  92.307692
```

Pull out spikes and check

```
#Show me the data that we have from the calculations
```

```
head(sampledat)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 1_202211_MSM_UP_LysA_10cm 264.6481 3270.8644 8.254775 92.266979
## 2 10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636 41.341089 193.951018
## 3 10_202211_MSM_TR_LysA_20cm_spk 1248.1115 6395.9330 38.930490 180.421241
## 4 100_202211_GCrew_TR_LysC_45cm 19.3955 133.2425 0.604975 3.758604
## 5 100_202211_GCrew_TR_LysC_45cm_spk 158.7476 134.0034 4.951578 3.780068
## 6 101_202211_GCrew_WC_SipA_10cm 705.7256 6762.9137 22.012651 190.773306
##      salinity
## 1 5.9104780
## 2 12.4241694
## 3 11.5574769
## 4 0.2407952
## 5 0.2421701
## 6 12.2206111
```

```
#pull out any rows that have "spk" in the SampleID column
```

```
spks <- sampledat %>%
```

```
  filter(str_detect(Sample_ID, "spk")) #have to change this to match data
```

```
head(spks)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 10_202211_MSM_TR_LysA_20cm_spk 1248.1115 6395.9330 38.930490 180.421241
## 2 100_202211_GCrew_TR_LysC_45cm_spk 158.7476 134.0034 4.951578 3.780068
## 3 110_202211_GCrew_SW_A_spk 956.5879 5163.2391 29.837427 145.648494
## 4 120_202211_GWI_TR_LysA_10cm_spk 1664.3307 12466.9403 51.912998 351.676736
## 5 130_202211_GWI_WC_SipA_20cm_spk 1900.5139 13145.0448 59.279910 370.805213
## 6 140_202211_GWI_RHZ_TR_SF_1_spk 1606.2517 11339.0519 50.101425 319.860420
##      salinity
## 1 11.5574769
## 2 0.2421701
## 3 9.3299991
## 4 22.5277871
## 5 23.7531220
## 6 20.4896928
```

```
#remove the dup from these IDs so we will have duplicate sample names
```

```
spks$Sample_ID<-gsub("_spk","",as.character(spks$Sample_ID))
```

```
spks <- spks[ , -c(2,3, 5,6)]
```

```
colnames(spks) <- c('Sample_ID', 'S04_mM_spk')
```

```
head(spks)
```

```
##           Sample_ID S04_mM_spk
## 1 10_202211_MSM_TR_LysA_20cm 38.930490
## 2 100_202211_GCrew_TR_LysC_45cm 4.951578
## 3 110_202211_GCrew_SW_A 29.837427
## 4 120_202211_GWI_TR_LysA_10cm 51.912998
## 5 130_202211_GWI_WC_SipA_20cm 59.279910
## 6 140_202211_GWI_RHZ_TR_SF_1 50.101425
```

```
#put it back together with the old data set and look for duplicates
QAspks <- merge(sampledats, spks)
head(QAspks)
```

```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1  10_202211_MSM_TR_LysA_20cm 1325.3953  6875.5636 41.341089 193.951018
## 2 100_202211_GCrew_TR_LysC_45cm   19.3955   133.2425  0.604975  3.758604
## 3    110_202211_GCrew_SW_A   663.6499  5151.5591 20.700246 145.319016
## 4   120_202211_GWI_TR_LysA_10cm 1862.6278 14578.7194 58.098185 411.247374
## 5   130_202211_GWI_WC_SipA_20cm 1879.0113 13943.5739 58.609211 393.330717
## 6   140_202211_GWI_RHZ_TR_SF_1 1641.0884 12090.9400 51.188035 341.070240
##      salinity S04_mM_spk
## 1 12.4241694  38.930490
## 2  0.2407952   4.951578
## 3  9.3088933  29.837427
## 4 26.3437720  51.912998
## 5 25.1960640  59.279910
## 6 21.8483546  50.101425
```

```
#now we need to calculate the spike concentration and calculate the spike recovery
spkconc <- (250/smw)      # in mM
spkvol <- 10              # in uL
spkvol <- spkvol/1000000
#spike for these samples was 10uL of the 250mM standard
QAspks$S04_spk_Conc <- (spkconc)*spkvol      # mmoles of S04
head(QAspks)
```

```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1  10_202211_MSM_TR_LysA_20cm 1325.3953  6875.5636 41.341089 193.951018
## 2 100_202211_GCrew_TR_LysC_45cm   19.3955   133.2425  0.604975  3.758604
## 3    110_202211_GCrew_SW_A   663.6499  5151.5591 20.700246 145.319016
## 4   120_202211_GWI_TR_LysA_10cm 1862.6278 14578.7194 58.098185 411.247374
## 5   130_202211_GWI_WC_SipA_20cm 1879.0113 13943.5739 58.609211 393.330717
## 6   140_202211_GWI_RHZ_TR_SF_1 1641.0884 12090.9400 51.188035 341.070240
##      salinity S04_mM_spk S04_spk_Conc
## 1 12.4241694  38.930490 7.797879e-05
## 2  0.2407952   4.951578 7.797879e-05
## 3  9.3088933  29.837427 7.797879e-05
## 4 26.3437720  51.912998 7.797879e-05
## 5 25.1960640  59.279910 7.797879e-05
## 6 21.8483546  50.101425 7.797879e-05
```

```
#need to determine dilution factors and initial amount of sample added
#if your samples are all the same dilution just use the first line
#for Steph / COMPASS this depends on the site so...
QAspks$Dilution <- 1
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 100, QAspks$Dilution)
```

```

QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 200, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$Dilution)
#QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$Dilution)

#Set Sample volumes in uL
QAspks$SampleVol <- 1
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 1501, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 1475, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 1462, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$SampleVol)
#QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$SampleVol)

#change sample volume to L
QAspks$SampleVol <- QAspks$SampleVol/1000000
head(QAspks)

```

```

##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1  10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636 41.341089 193.951018
## 2 100_202211_GCrew_TR_LysC_45cm  19.3955  133.2425  0.604975  3.758604
## 3   110_202211_GCrew_SW_A 663.6499 5151.5591 20.700246 145.319016
## 4  120_202211_GWI_TR_LysA_10cm 1862.6278 14578.7194 58.098185 411.247374
## 5  130_202211_GWI_WC_SipA_20cm 1879.0113 13943.5739 58.609211 393.330717
## 6  140_202211_GWI_RHZ_TR_SF_1 1641.0884 12090.9400 51.188035 341.070240
##      salinity S04_mM_spk S04_spk_Conc Dilution SampleVol
## 1 12.4241694  38.930490 7.797879e-05      50  0.001501
## 2  0.2407952   4.951578 7.797879e-05      50  0.001501
## 3  9.3088933  29.837427 7.797879e-05       1  0.000001
## 4 26.3437720 51.912998 7.797879e-05     100  0.001475
## 5 25.1960640 59.279910 7.797879e-05     200  0.001462
## 6 21.8483546 50.101425 7.797879e-05       1  0.000001

```

```

#gives us the total SO4 in the sample in mmoles
QAspks$S04_Total_unspkd <- (QAspks$S04_mM/QAspks$Dilution)*(QAspks$SampleVol)

##total SO4 in spiked sample in mmoles
QAspks$S04_Total_spkd <- (QAspks$S04_mM_spk/QAspks$Dilution)*(QAspks$SampleVol+spkvol)

QAspks$S04_expctd_spkd <- (QAspks$S04_Total_unspkd + QAspks$S04_spk_Conc)
QAspks$spk_recovery <- (QAspks$S04_Total_spkd/QAspks$S04_expctd_spkd)*100
QAspks$S04_spks_flag <- ifelse(QAspks$spk_recovery <=120 & QAspks$spk_recovery >=80 , 'YES', 'NO, rerun')

head(QAspks)

```

```
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1    10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636 41.341089 193.951018
## 2 100_202211_GCrew_TR_LysC_45cm   19.3955   133.2425  0.604975   3.758604
## 3    110_202211_GCrew_SW_A    663.6499 5151.5591 20.700246 145.319016
## 4    120_202211_GWI_TR_LysA_10cm 1862.6278 14578.7194 58.098185 411.247374
## 5    130_202211_GWI_WC_SipA_20cm 1879.0113 13943.5739 58.609211 393.330717
## 6    140_202211_GWI_RHZ_TR_SF_1 1641.0884 12090.9400 51.188035 341.070240
##   salinity S04_mM_spk S04_spk_Conc Dilution SampleVol S04_Total_unspkd
## 1 12.4241694 38.930490 7.797879e-05      50 0.001501 1.241059e-03
## 2  0.2407952  4.951578 7.797879e-05      50 0.001501 1.816135e-05
## 3  9.3088933 29.837427 7.797879e-05       1 0.000001 2.070025e-05
## 4 26.3437720 51.912998 7.797879e-05     100 0.001475 8.569482e-04
## 5 25.1960640 59.279910 7.797879e-05     200 0.001462 4.284333e-04
## 6 21.8483546 50.101425 7.797879e-05       1 0.000001 5.118803e-05
##   S04_Total_spkd S04_expctd_spkd spk_recovery S04_spks_flag
## 1 0.0011764794 1.319038e-03 89.19221      YES
## 2 0.0001496367 9.614014e-05 155.64435     NO, rerun
## 3 0.0003282117 9.867904e-05 332.60529     NO, rerun
## 4 0.0007709080 9.349270e-04 82.45649      YES
## 5 0.0004363001 5.064121e-04 86.15515      YES
## 6 0.0005511157 1.291668e-04 426.66968     NO, rerun
```

#plot spk recoveries output as a bar graph to easily check - want any over 10% to be red need to work on

```
spksbar <- ggplot(data = QAspks, aes(x = Sample_ID, y = spk_recovery)) +
  geom_bar(stat = 'identity') +
  scale_fill_manual(values=c("lightblue4")) +
  theme_classic() + labs(x= "Sample ID", y="Spike Recovery (%)") +
  theme(legend.position="none") +
  geom_hline(yintercept=80, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=120, linetype="dashed", color = "black", size=1)
```

```
spksbar
```



```
colnames(IDs) <- c("Analysis_No" , "Date","Site", "Zone", "Replicate", "Depth", "Tree")
head(IDs)
```

```
##   Analysis_No   Date   Site Zone Replicate Depth Tree
## 1           1 202211   MSM  UP      LysA   10cm    1
## 2          10 202211   MSM  TR      LysA   20cm   10
## 3         100 202211 GCrew  TR      LysC   45cm  100
## 4         101 202211 GCrew  WC      SipA   10cm  101
## 5         102 202211 GCrew  WC      SipA   20cm  102
## 6         103 202211 GCrew  WC      SipA   45cm  103
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, sampled2)
head(alldat)
```

```
##   Analysis_No   Date   Site Zone Replicate Depth Tree
## 1           1 202211   MSM  UP      LysA   10cm    1
## 2          10 202211   MSM  TR      LysA   20cm   10
## 3         100 202211 GCrew  TR      LysC   45cm  100
## 4         101 202211 GCrew  WC      SipA   10cm  101
## 5         102 202211 GCrew  WC      SipA   20cm  102
## 6         103 202211 GCrew  WC      SipA   45cm  103
##           Sample_ID   S04_ppm   Cl_ppm   S04_mM   Cl_mM
## 1  1_202211_MSM_UP_LysA_10cm 264.6481 3270.8644 8.254775 92.266979
## 2  10_202211_MSM_TR_LysA_20cm 1325.3953 6875.5636 41.341089 193.951018
## 3 100_202211_GCrew_TR_LysC_45cm  19.3955  133.2425  0.604975  3.758604
## 4 101_202211_GCrew_WC_SipA_10cm 705.7256 6762.9137 22.012651 190.773306
## 5 102_202211_GCrew_WC_SipA_20cm 241.4455 6041.3935  7.531051 170.420127
## 6 103_202211_GCrew_WC_SipA_45cm 630.0428 6834.2797 19.651990 192.786451
##   salinity
## 1 5.9104780
## 2 12.4241694
## 3 0.2407952
## 4 12.2206111
## 5 10.9168241
## 6 12.3495694
```

Make final dataframe with IDs

```
write.csv(alldat, file="Processed Data/COMPASS_SynopticCB_PW_Processed_Cl_S04_202211.csv") #Change fil
```

END